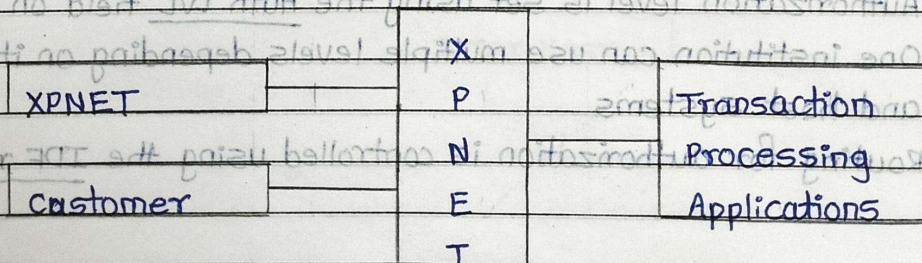


XPNET Overview

* XPNET System:-

- An XPNET system provides the network management layer for BASE24 transaction processing
- Serves as the software platform on which the application process run
- Provides functions for configuration and management of the network and its components
- The XPNET is the main process installed on Tandem for BASE24 system under which there are processes like DH, Auth and HISO.
- It is the main process which manages the transaction processing in the BASE24.
- It handles the different messages which need to be passed to the host bank.
- It manages the network configuration data like IP addresses and port numbers.
- It manages a log file which can be audited afterwards.
- It handles any misleading/mishandling of the system or any software or hardware failure in the transaction processing.
- It checks if the connection is a socket communication or a inter-process communication or moreover it checks the connection lines and the protocols that are to be used.



★ XPNET components:Tandem SystemXPNET systemLogical Network 1LN2LN3

DB

XPNET Node

DB

database

DB

Tandem System:

- Tandem (also called as HP Nonstop Server originally developed by Tandem computers, is a series of fault tolerant systems of server (Hardware) designed for mission-critical applications; known for their high availability and scalability.
- The physical tandem computer:

Tandem

\system

A

\system

B

\system

C

\system

D

★ XPNET System:

- It is a structural network of one or more Logical Networks.

Logical Network:

- One or more XPNET nodes, acting as a single settlement entity, accessing a single database.

XPNET Node:

- A Node is a grouping of one or more stations.
- Represents a logical unit within the communication network.
- Helps in managing and organizing stations for routing and processing.
- Often used to represent a branch or regional group of terminals.
- Example: All ATMs in one city might be under a single node.
- It consists of an xpnnet process and the objects that it manages.
- These objects are nodes, lines, stations, application processes and links to other nodes.

XPNET Node

Station 1	line	XPNET Processes	Application Processes
Station 2	line	XPNET Processes	
Station 3	line		PIA^Link^PIB

Station:

- A station is the actual endpoint device like an individual ATM, POS terminal, or even a host system.
- Each station has a unique ID.
- Communicates over a line belongs to a node.
- Example: A specific ATM Located at a bank branch

Line:

- A line represents a physical or logical communication path
- It connects the BASE24 system to a remote device (like ATM or POS) or a host system.
- Can be over TCP/IP, leased lines, dial up, etc.
- Managed by a Line Driver (LD) in XPNET.
- Example: A TCP/IP link between BASE24 and a remote ATM

Hierarchy Overview:-

Line → carries communication data



Node → Logical grouping of devices



Station → Actual terminal or host endpoint

* XPNET component Names:

- Objects and files can be given any valid name
- Use of ACI naming conventions is not required, but is highly recommended to:
 - o Help enforce uniqueness of object and file names
 - o Provide a point of reference for discussing system components
- (HELP24)

XPNET:- For XPNET NETWORK
 NCP.COM → (commandline interface)
 NETARD NETDMPRD
 NETCJRD

TESTCNTL:-
 GOAFT GOENTL GONCP GOPRE LICONE PATHCOLD
 PATHCONF PATHCOOL PATHCTL
 PATHSTOP PATHSTRT PRECMD
 SHUTDOWN STRTLOG

TESTDATA:-
 - NIANEF NCSP NCSP7
 - NCSS NMAP PREMAP
 - SPROUTE

TESIDATA: Application Files

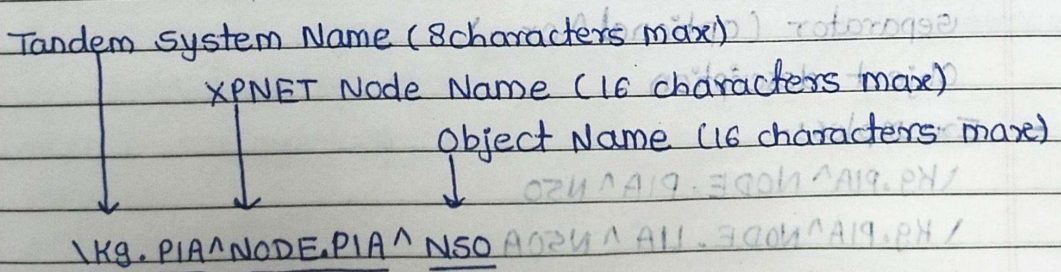
TESITPLT :- Application File Templates

TES-PIANCJA:- CJO0000 CJO0001

PIANNOMA :- NOM00000 NOM00001

* Fully Qualified Object Name:-

- Fully qualified XPNET object names consists of:



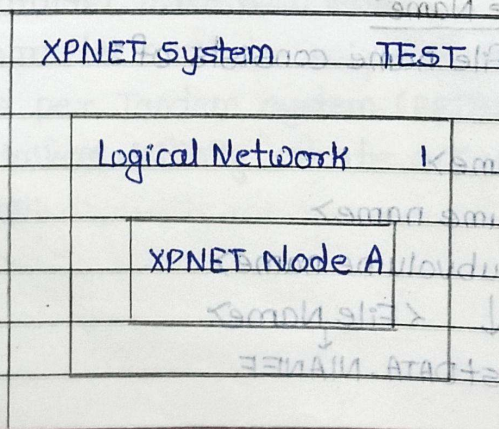
Logical Network Configuration File (LNCF):-

- contains file assignments and processing parameters for the logical network
- Assigns relate file names to the physical location to that file
- Assign: SEC
To : \Kg. \$DATA\AENV\DATA\SEC
- Params provide configurable processing parameters for the Logical network
- PARAM:
Name: Logical-NET
Text : AENI

Network Environment File:

- Objects that make up the XPNET nodes are defined in the network Environment File (NEF) → database file of node
- Node: PIA^Node
- Processes: PIA^NSO
- Links : PIA^LINK^PIB
- Devices : Dev^NSO
- Lines : LIA^NSOA
- Stations : GIANSOAI

★# XPNET File Structure:



Subvolume Names:

- Identify subvolumes on which various files reside
- Part of the fully qualified file name
- Upto 8 alphanumeric characters
- May include:-
 - o XPNET System name
 - o Logical network identifier

Examples:

TESTCNTL

TESTDATA (Network)

TESIDATA (Application)

TESITPLT

File Name:

- Uniquely identifies the files defined to the system
- May include:-
 - o Logical network identifier
 - o XPNET node identifier
 - o File identifier/ sequence number.

Examples:-

LICONF

NIANEF

CJO0000

Fully Qualified File Name:

- A fully qualified file name consists of:

<Tandem System name>

<Disk volume name>

<Subvolume name>

<File Name>

\Kg. \$DATA1.TestDATA.NIANEF

Tandem System Name:

- Identifies the tandem system
- Part of the fully qualified file name
- First character must be a backslash
- Up to 8 characters (including the backslash)
- Example:

\Tandem

\5451

\Kg

Volume Name:

- Identify disk volumes on which the system files resides
- Part of the fully qualified system file name.
- First character must be \$, followed by an alphabetic character.
- Up to 8 characters (including the \$)
- Example:-

\$ system

\$ Bank

\$ DATA

XPNET System Name:

- Describes an entire XPNET system environment
- Typically four alpha characters:
- First character must be a letter
- First character must be unique if there will be multiple XPNET systems per Tandem system (PPDNames)
- ACI installers initially create a Test system.

Examples:

TEST

PROD

DVLP

CERT

- Failover Behaviour:

Seamless takeover, no restart of processes or links

- Use Cases:

High availability, zero message loss environments.

o Warm Backup:

- Message Checkpointing:

only at initialization, messages in progress are lost on failure

- sync Frequency:

One-time sync at startup

- Failure Recovery:

~ ADOPT OFF: All stations/processes/links are restarted

~ ADOPT ON: Connectivity is re-established without restart

- Failover Behaviour:-

Involves restart unless associative

- Use Cases:-

Medium-critical systems where minor message loss is acceptable

o Object Machine States:

- An XPNET object may be in one of seven states defined by its readiness to perform its functions:

1. starting
2. started
3. stopping (not links)
4. stopped
5. Abnormal
6. Suspended (stations only)
7. Disabled

o Object Startup:-

- XPNET manages the starting of the objects in the XPNET Node
- XPNET can start object based on the following table:

Object Type	Automatic	Command (default)	Demand
Node	X	X	
Process	X	X	X
LINK	X	X	

Startup Attribute

o Object Recovery:

- The XPNET process can be configured to provide fault tolerant processing by:
 - o Existing on multiple CPUs
 - o Running multiple copies of critical application processes on different CPUs
- XPNET process checkpoints with its backup in one of two ways
 - o Hot
 - o Warm
 - o Hot Backup:

- Message checkpointing:-

All in-progress messages are checkpointed

- Sync Frequency:

continuous syncing of messages and config changes.

- Failure Recovery:-

No message loss; backup resumes in-progress messages

★ PPD Name:

- PPD names identify a process pair
- Name must be unique on the Tandem System.
- First character must be a \$, followed by upto 4 characters
- May include:
 - First character of XPNET system name
 - Logical network identifier
 - XPNET node identifier
 - Copy number.
- Examples:
 - \$TIAN
 - \$TPMN
 - \$TIAU
 - \$TATI

★ XPNET Functions:-

XPNET Process:

- The controlling process for the XPNET node responsible for:
 - o Driving the data communication lines
 - o Controlling all lines, stations and processes in the XPNET node
 - o Routing messages
 - o Managing message queues
 - o Running system audits
 - o Communicating with other XPNET processes
 - o Monitoring the performances of the System
 - o Enabling continuous processing.

Object Management:

- The XPNET process manages the objects in the XPNET node, as defined in the NEF
- object management:
 - o Startup
 - o Recovery
 - o States

Tandem System Name:-

- Identifies the Tandem system
- Part of fully qualified object name
- First character must be a backslash
- Upto 8 characters (including backslash)
- Examples:-

\TANDEM

\SYS1

\Kg

XPNET Node Name:-

- Identifies the XPNET Node
- Part of fully qualified object name
- Upto 16 alphanumeric characters
- Examples:-

\Kg.PIA^NODE

\Kg.PIB^NODE

\Kg.PIC^NODE

Object Name:-

- A symbolic name that uniquely identifies each object in the XPNET Node.
- Upto 16 characters (alphanumeric)
- May include:
 - Object type indicator
 - Logical network identifier
 - XPNET node identifier
 - Separator (optional ^ or -)
 - Object description

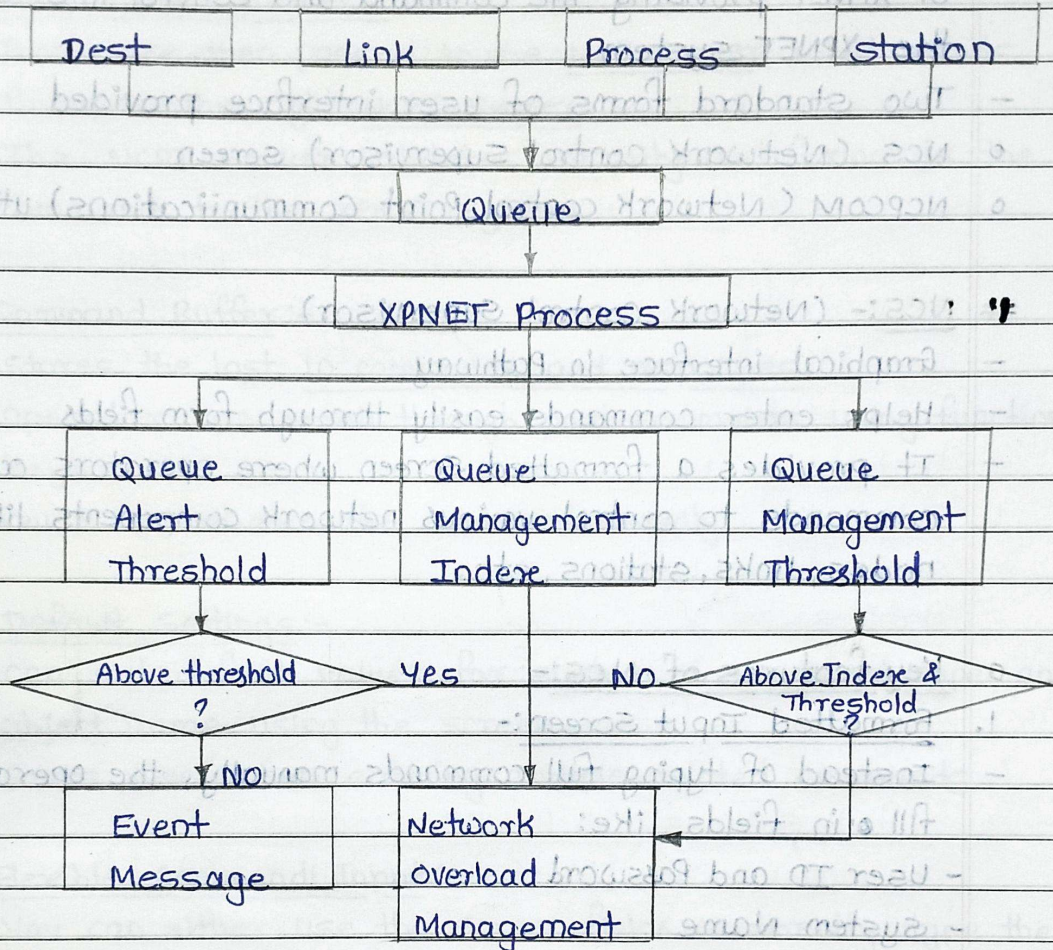
Examples:-

\Kg.PIA^NODE.PIA^NSO

\Kg.PIA^NODE.LIA^NSOA

\Kg.PIA^NODE.SIAN50AI

★ Queue Facilities:-



★ Command and Control:

- Operator commands provide the syntax for operators to modify or query the system

- Three types of operator commands:

- o Environment commands
- o ~~Environ~~ Inquiry commands
- o Object management commands.

★ Network Overload Management (NOM) Actions:

For each message evaluated during overload:

- If marked as "Do Not Discard" → it is queued
- If marked as "Return" → it is returned to sender
- If Saving NOM Messages to disk is enabled → message saved to extended memory
- otherwise → message is discarded.
- Event messages are generated periodically with discard counts
- Applications must set flags in message headers for discard, return, etc.
- Also:
 - o - If message is from a station with NOMOVERRIDE = ON, it is saved instead of discarded.
 - o - Once memory or message count drops below thresholds, overload actions stop.

★ Application Participation:

- Message headers include QAT% and NOM status.
- Apps can intelligently manage their traffic (e.g. stop sending low-priority messages).

★ Event Logging:

- Provides logging of events generated by XPNET or of its objects
- o Used in detecting and resolving system problems.

- o Service based: Message routed based on the service that it requires
- Service = Auth

* Message Queue Management:

- Provides a means of monitoring and managing messages queued to objects.

User Configuration limits:

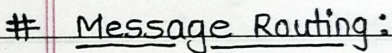
- Queue Alert Threshold
- Queue Management Threshold
- Queue Management Index

1. Queue Alert Threshold (QAT)

- A user-defined message count that, when exceeded, triggers event messages
- A timer starts: when it expires, the queue is re-evaluated
- If still over the threshold → another alert;
if below → a recovery event is generated.
- The current QAT percentage is included in message header to help applications take proactive action (eg. drop requests, stand in).
- Messages are sent to the RTS clone process if enabled.

2. Queue Management Index (QMI) and Queue Management Threshold (QMT):

- QMI: % of total XPNET memory used
- QMT: number of messages in the queue
- When both are exceeded, the XPNET process:
Generates event Messages
Starts Network Overload Management (NOM)



- The XPNET process routes messages to:
 - o Application processes in the XPNET node
 - o Other XPNET Nodes
 - o Stations outside of the XPNET system
- Hosts
- Interchanges
- Terminals
- other endpoints

Types of Message Routing:

- o Symbolic:- Messages routed based on symbolic name
- Destination = PIA^NSOA

flow



4. Command Execution without Pathway:

- You type the command in NCP.COM
- NCP.COM validates syntax
- Sends the command to XPMON process
- XPMON forwards it to the NCPI server (via NCP server if needed)
- The NCPI server validates security and processes the command.

* Network Control (XPNET) - short Notes:

- Network control in XPNET means managing and monitoring the network to ensure everything (nodes, links, processes, stations) runs smoothly and efficiently.

Key Functions:-

- Monitor Network operations
- Start/stop processes, nodes, links
- Reconfigure devices and destinations.
- View and manage logs, events and security

Main components:

o Operator Interfaces:

- Used to issue commands to XPNET System.
- Includes:

NCP.COM Utility: Command line interface

NCS screen : Menu-driven screen interface (used only with Pathway)

Custom Interfaces: Created by users or third parties

o Server Processes:

- XPMON: Main control point for commands (non-Pathway)
- NCPI server: Handles security and sends commands to XPNET.

o How to start NCP.COM?

- You start NCP.COM by entering the RUN command from the TACL prompt.
- The first command you must enter in an NCP.COM session is:
 - Path :- If using Pathway environment
 - Gate :- If Pathway is not licensed (used to connect to XPMON instead)
- If security is enabled, you must also log in with a valid user ID and Password.

o Features of NCP.COM:

1. Interactive Prompt:

- Shows prompts like:

NODE 8>

(Here, NODE is the default object type, and 8 is the command count)

2. Default Settings with ASSUME Command:

- You can use the ASSUME command to set default values for:
 - system
 - Node
 - object type
 - Object Name
- Reduces typing when issuing multiple commands for the same object.

3. Command History:

- Maintains a buffer of the last 20 commands (but not their responses)
- Use the "HISTORY" command to view past commands and re-execute them.

3. Command Processing:

- commands from the NCS screen are interpreted by the NCS Server process
- These are then passed to the NCP server, which sends them to the right NCPI Server.
- The NCPI Server validates security and forwards the command to the XPNET System.

4. Command Buffer:-

- Stores the last 10 commands and responses.
- Operator can scroll through past commands using function keys.
- Reuse or re-execute commands easily!

5. Default Settings:-

- can set default values for system, node, object type, and object name using the screen.
- Saves time when entering multiple related commands.

6. Flexible Command Input:-

- You can either use the screen fields or directly type the full command in the command keyword line

NCPCOM:- (Network Control Point Communications)

- NCPCOM is a command-line utility used to manage and control the XPNET network
- It provides an interactive prompt where the network operator can type and execute network control commands.
- It is available in both Pathway and non-pathway environments

★ command and control Facilities:

- Open Network Control Facility (ONCF) is the component of XPNET providing the command and control interface to the XPNET system.
- Two standard forms of user interface provided
 - o NCS (Network Control Supervisor) screen
 - o NCPcom (Network control Point Communications) utility

NCS:- (Network Control Supervisor)

- Graphical interface in Pathway
- Helps enter commands easily through form fields
- It provides a formatted screen where operators can enter commands to control various network components like nodes, links, stations, etc.

o Key features of NCS:-

1. Formatted Input Screen:-

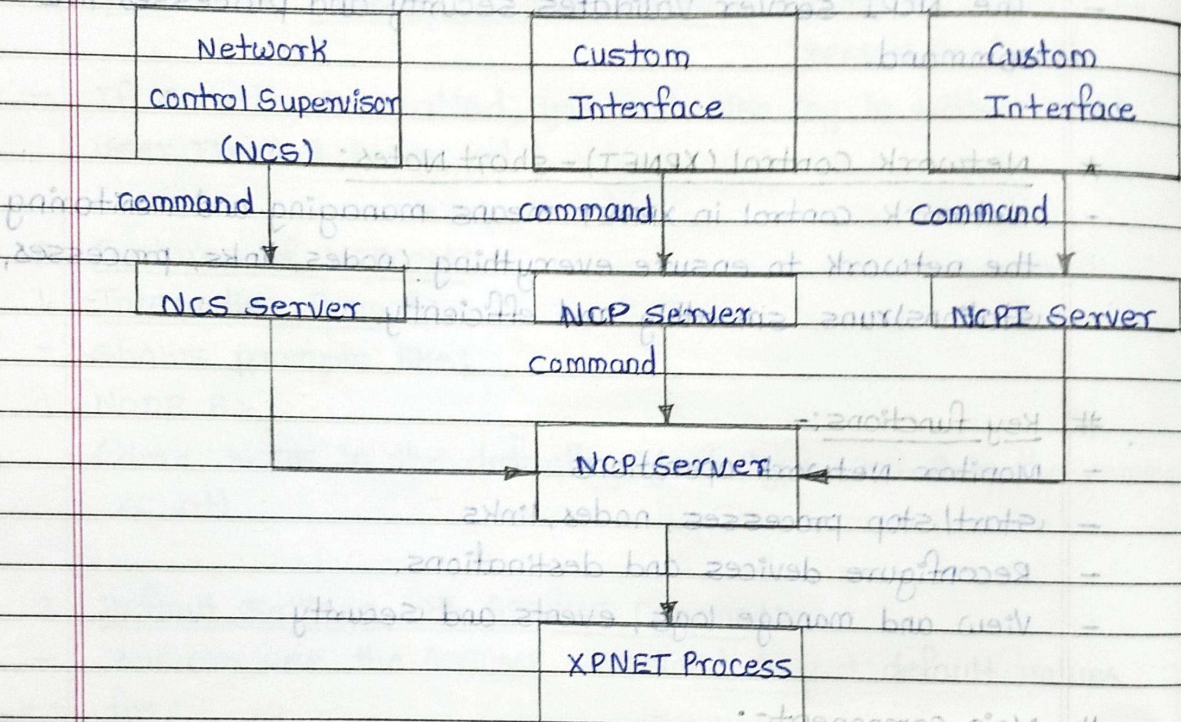
- Instead of typing full commands manually, the operator can fill in fields like:
 - User ID and Password
 - System Name
 - XPNET Node Name
 - Object type (eg. Node, Link, Station)
 - Object Name
 - Command Keyword
- Once filled, pressing a function key sends the command for execution.

2. Easy to use:-

- Helps users by guiding them through menus and input fields.
- Reduces command syntax errors.

- NCP Server: Finds the correct node and forwards commands to NCPI
- NCS Server: Used with NCS screen in Pathway to interpret commands.

Network Control in a Pathway Environment:



* Network control security:

- XPNET provides security measures to limit access by command and to optionally audit commands initiated
 - o By node
 - o By operator
 - o By command

* Message Auditing

→ used to detect and diagnose problems in transaction processing.

- XPNET provides an auditing facility for tracing message traffic between XPNET process and designated stations and processes.